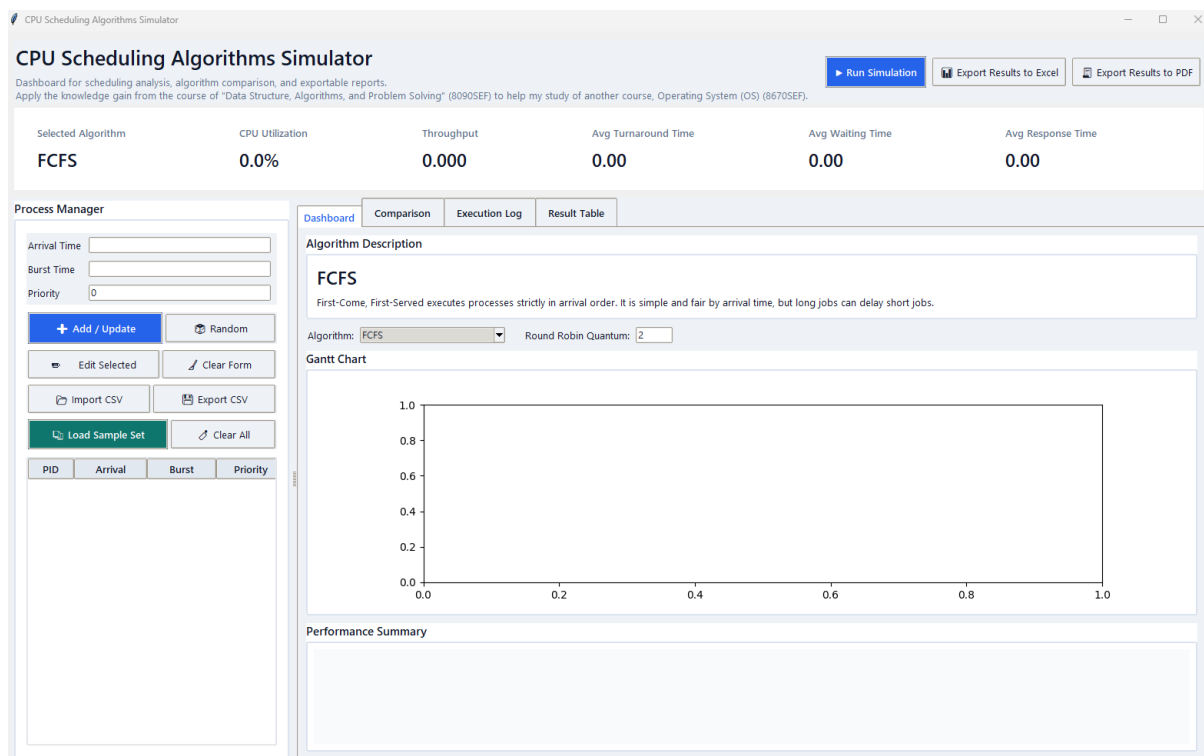


Appendix A — User Guide

1. Starting the CPU Scheduling Algorithms Simulator

(i)	Ensure Python 3.10 or later is installed.
(ii)	Install dependencies: <code>pip install matplotlib openpyxl pillow</code>
(iii)	Run: <code>python main.py</code>

The main application window



A. PROCESS DATA

1. Adding Processes

There are four ways to populate the process list:

(i)	Manual entry	Fill in Arrival Time, Burst Time, and Priority fields in the left panel. Arrival Time Burst Time Priority 0 Then click Add / Update.  Add / Update Click Clear Form to create a blank form for data entry, if necessary.  Clear Form
(ii)	Random	Click Random to add a process with randomly generated values.  Random
(iii)	Sample Set	Click Load Sample Set to load predefined processes.  Load Sample Set
(iv)	Import CSV	Click Import CSV and select a CSV file with columns Arrival Time, Burst Time, Priority.  Import CSV

2. Editing a process

- (i) Double click a row in the process list.

PID	Arrival	Burst	Priority
1	0.0	12	0
2	4.0	6	0
3	10.0	2	0

- (ii) Values populate the form fields.

Arrival Time	4.0
Burst Time	6
Priority	0

- (iii) Modify as needed and click **Add / Update** to apply changes.

Arrival Time	4.0
Burst Time	999
Priority	0

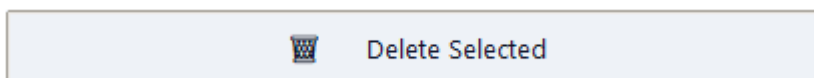
 Add / Update

3. Deleting a process

- (i) Select a row.

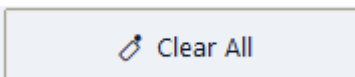
PID	Arrival	Burst	Priority
1	0.0	12	0
2	4.0	6	0
3	10.0	2	0

- (ii) Click **Delete Selected**.

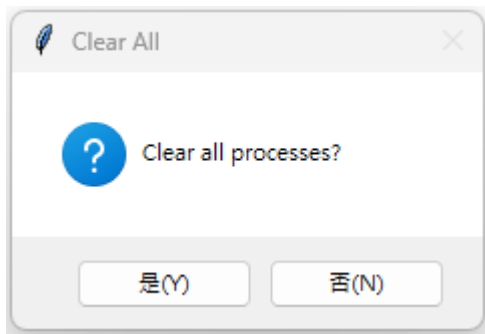


4. Clear All Processes

- (i) Click **Clear All** to initial the deletion all processes.



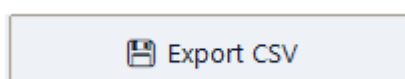
- (ii) A pop-up message will be displayed.



- (iii) Click **是(Y)** to delete all processes. Cancel deletion, please click **否(N)**.

5. Exporting the process list

Click **Export CSV** to save the current list to a CSV file for reuse.



B. RUNNING THE SIMULATOR

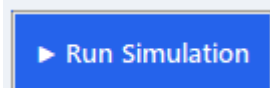
1. Ensure the process list contains at least one process.

PID	Arrival	Burst	Priority
1	0.0	12	0
2	4.0	6	0
3	10.0	2	0

2. If Round Robin results are needed, set the desired time quantum in the Round Robin Quantum field (default: 2).

Round Robin Quantum:

3. Click **Run Simulation** in the header bar.



4. All six algorithms execute simultaneously. The KPI strip at the top updates immediately.

Before:

Selected Algorithm	CPU Utilization	Throughput	Avg Turnaround Time	Avg Waiting Time	Avg Response Time
FCFS	0.0%	0.000	0.00	0.00	0.00

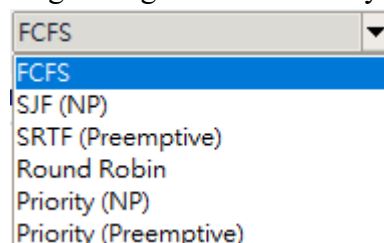
After:

Selected Algorithm	CPU Utilization	Throughput	Avg Turnaround Time	Avg Waiting Time	Avg Response Time
FCFS	100.0%	0.150	12.00	5.33	5.33

5. Select an algorithm from the Algorithm dropdown to switch the Dashboard view.

Algorithm:

The following six algorithms are ready for selection:



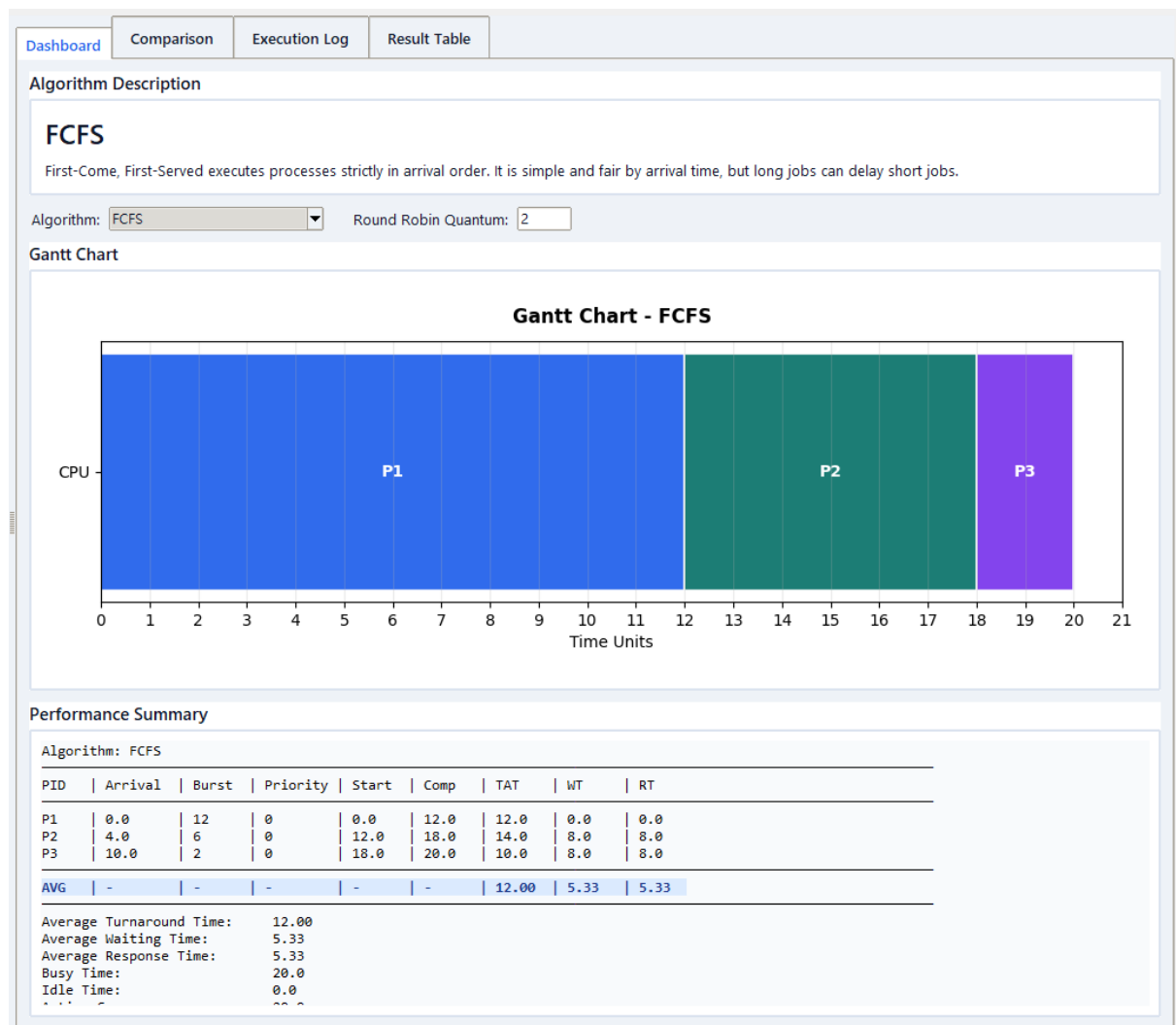
Running the simulation again with a modified process list or quantum value overwrites the previous results.

C. DATA ANALYSIS

1. Dashboard Tab

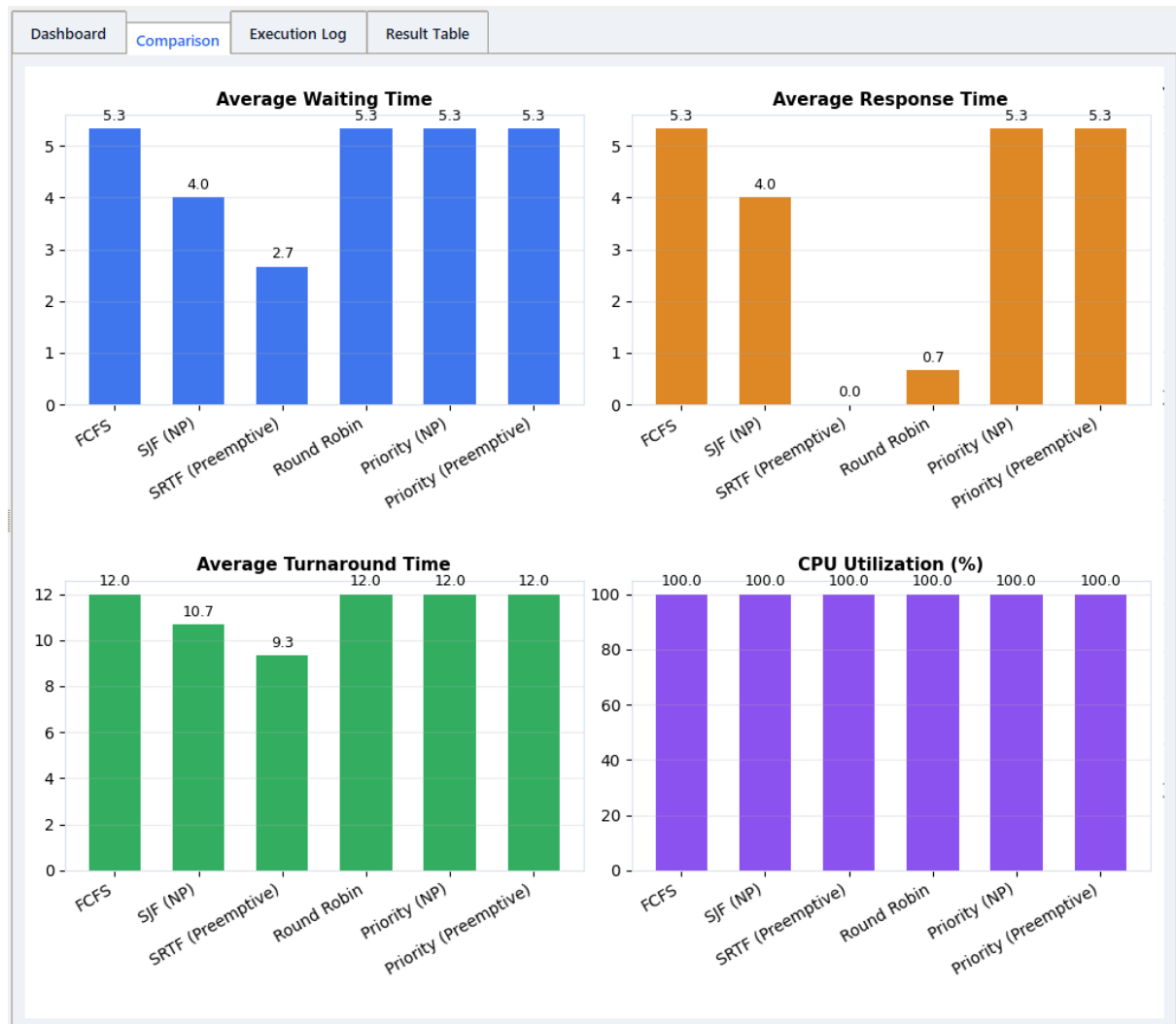
The Dashboard tab displays results for the algorithm currently selected in the dropdown:

(i)	Algorithm Description	Explanation of the selected algorithm
(ii)	Gantt Chart	Colored timeline bar showing which process occupied the CPU at each time unit.
(iii)	Performance Summary	A table listing per-process Start Time, Completion Time, Turnaround Time, Waiting Time, and Response Time, with an average row at the bottom, with summary of key indicators.



2. Comparison Tab

Four bar charts showed with Average Turnaround Time, Average Waiting Time, Average Response Time, and CPU Utilization, for immediate identification of which algorithm performs best on each metric for the current process set.



Lower bars in the TAT, WT, and RT charts indicate better scheduling efficiency.

A higher bar in the CPU Utilization chart indicates less idle time.

3. Execution Log Tab

The Execution Log is a useful study tool for tracing algorithm decisions step by step. The tab shows a timestamped, row-by-row trace of every scheduling event for the selected algorithm.

Event Type	Meaning
ARRIVED	A process joined the ready queue at this time unit
STARTED	The CPU was allocated to a process (first time)
PREEMPTED	The running process was interrupted.
COMPLETED	A process finished; TAT, WT, and RT values shown
IDLE	No process was ready. the CPU was idle this tick

Dashboard	Comparison	Execution Log	Result Table
Time	Event	Running	Ready Queue
0.0	Arrival	-	[1]
0.0	Execution Start	P1	[]
4.0	Arrival	-	[2]
4.0	Preemption	P1	[2, 1]
4.0	Execution Start	P2	[1]
10.0	Completion	P2	[1]
10.0	Arrival	-	[3]
10.0	Execution Start	P3	[1]
12.0	Completion	P3	[1]
12.0	Execution Start	P1	[]
20.0	Completion	P1	[]

4. Result Table Tab

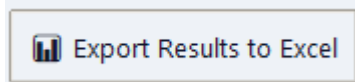
The Result Table tab shows a detailed per-process metrics table. The highlighted AVG row at the bottom shows the average of each metric across all processes.

Dashboard	Comparison	Execution Log	Result Table					
PID	Arrival	Burst	Priority	Start	Completion	TAT	WT	RT
1	0.0	12	0	0.0	20.0	20.0	8.0	0.0
2	4.0	6	0	4.0	10.0	6.0	0.0	0.0
3	10.0	2	0	10.0	12.0	2.0	0.0	0.0
AVG	-	-	-	-	-	9.33	2.67	0.00

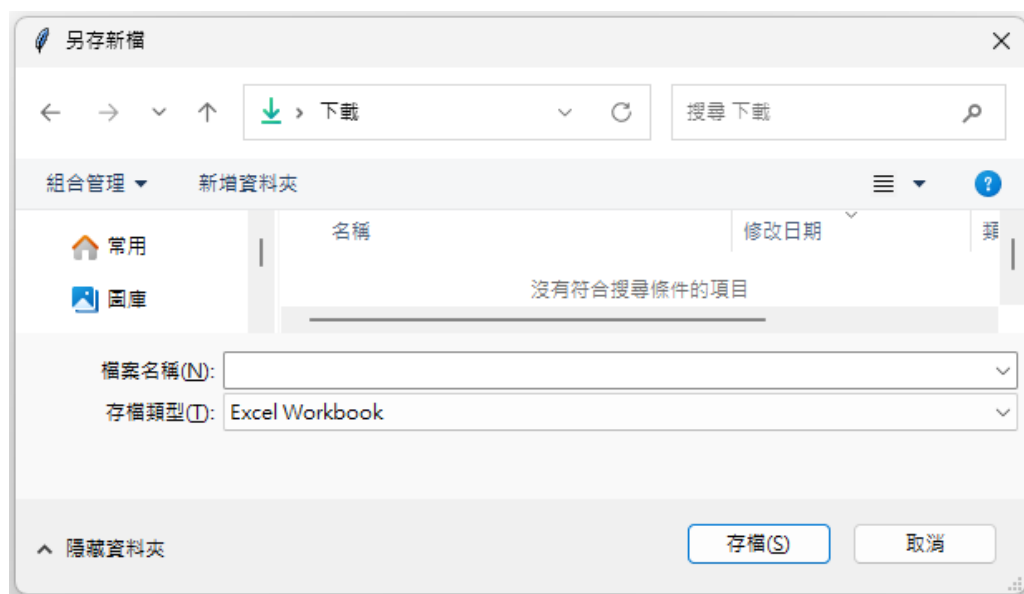
D. EXPORTING RESULTS

1. Export to Excel

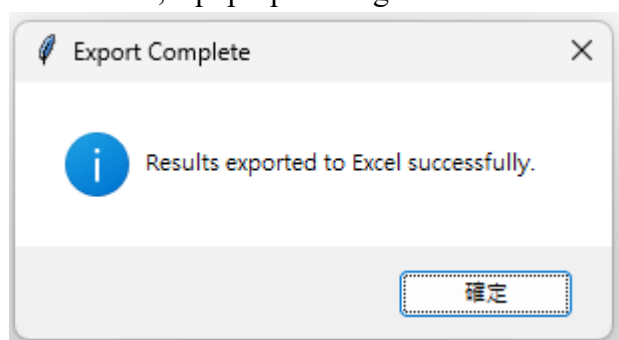
- (i) Click **Export Results to Excel** in the header bar.



- (ii) Choose a save location in the file dialog.

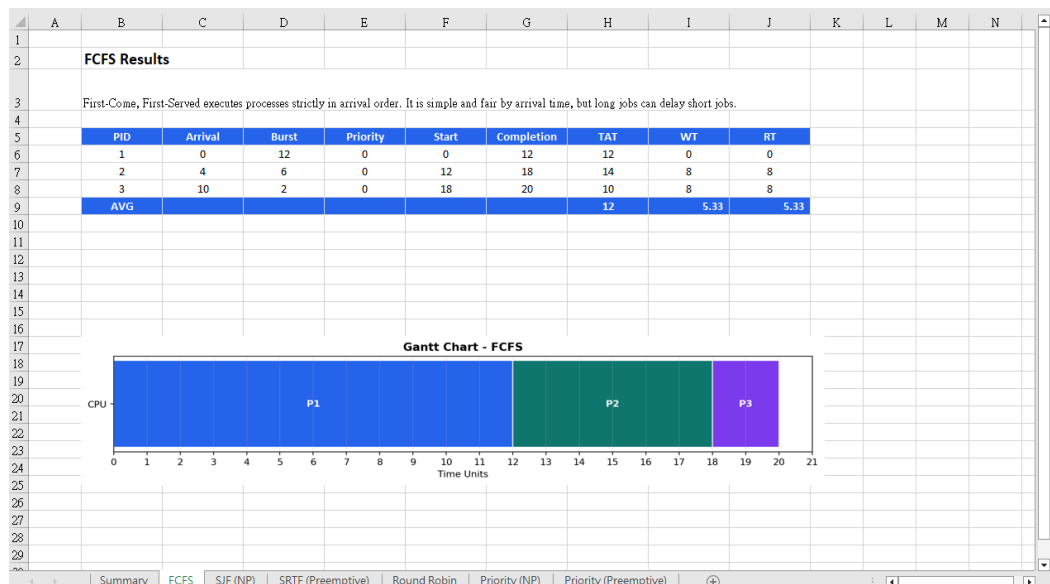
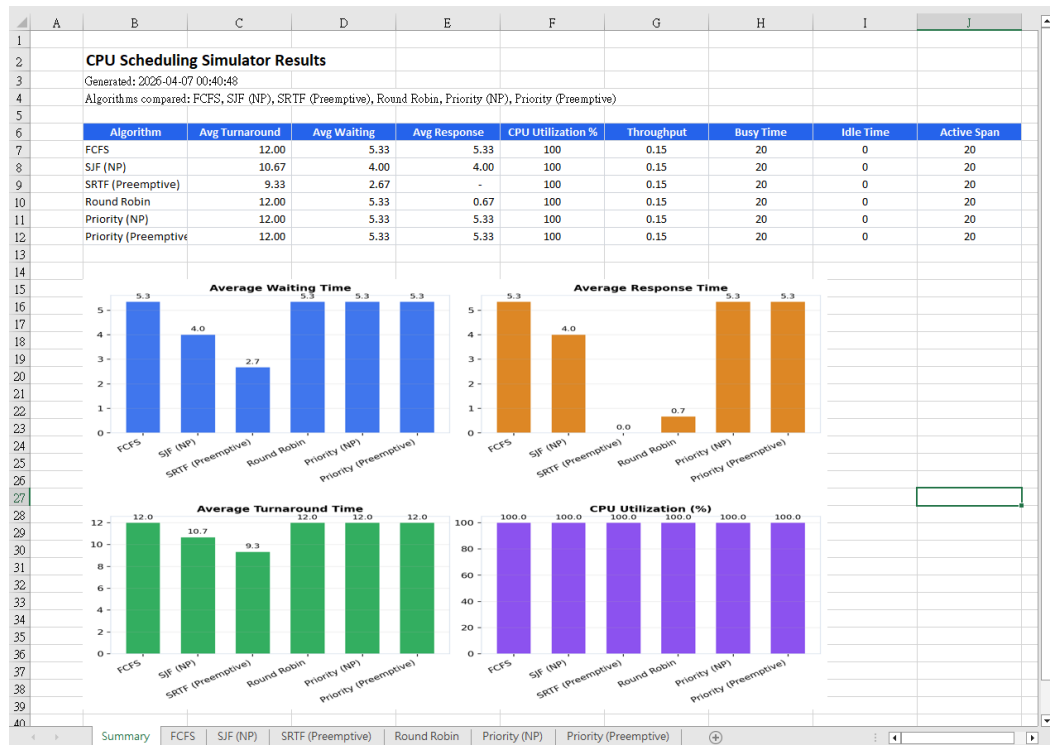


- (iii) Once saved, a pop-up message will be shown.



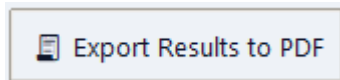
COMP8090SEF Individual Project

- (iv) The workbook contains:
- (a) One Summary sheet (comparison table + charts); and
 - (b) Six algorithm sheets (process results table + Gantt chart)

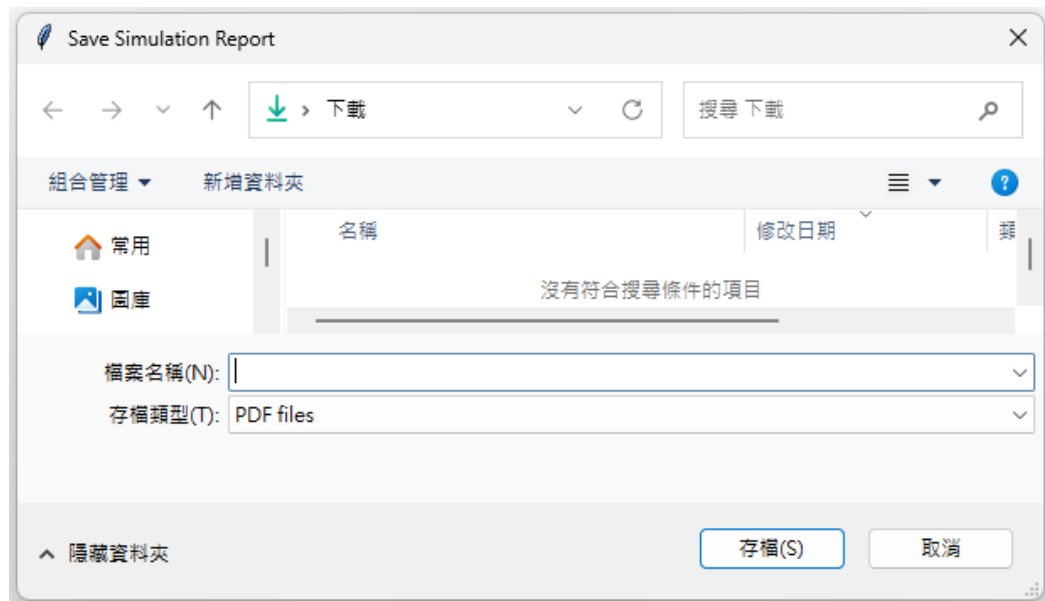


2. Export to PDF

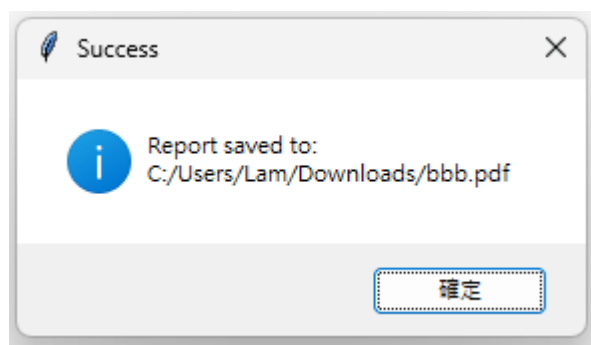
- (i) Click **Export Results to PDF** in the header bar.



- (ii) Choose a save location in the file dialog.

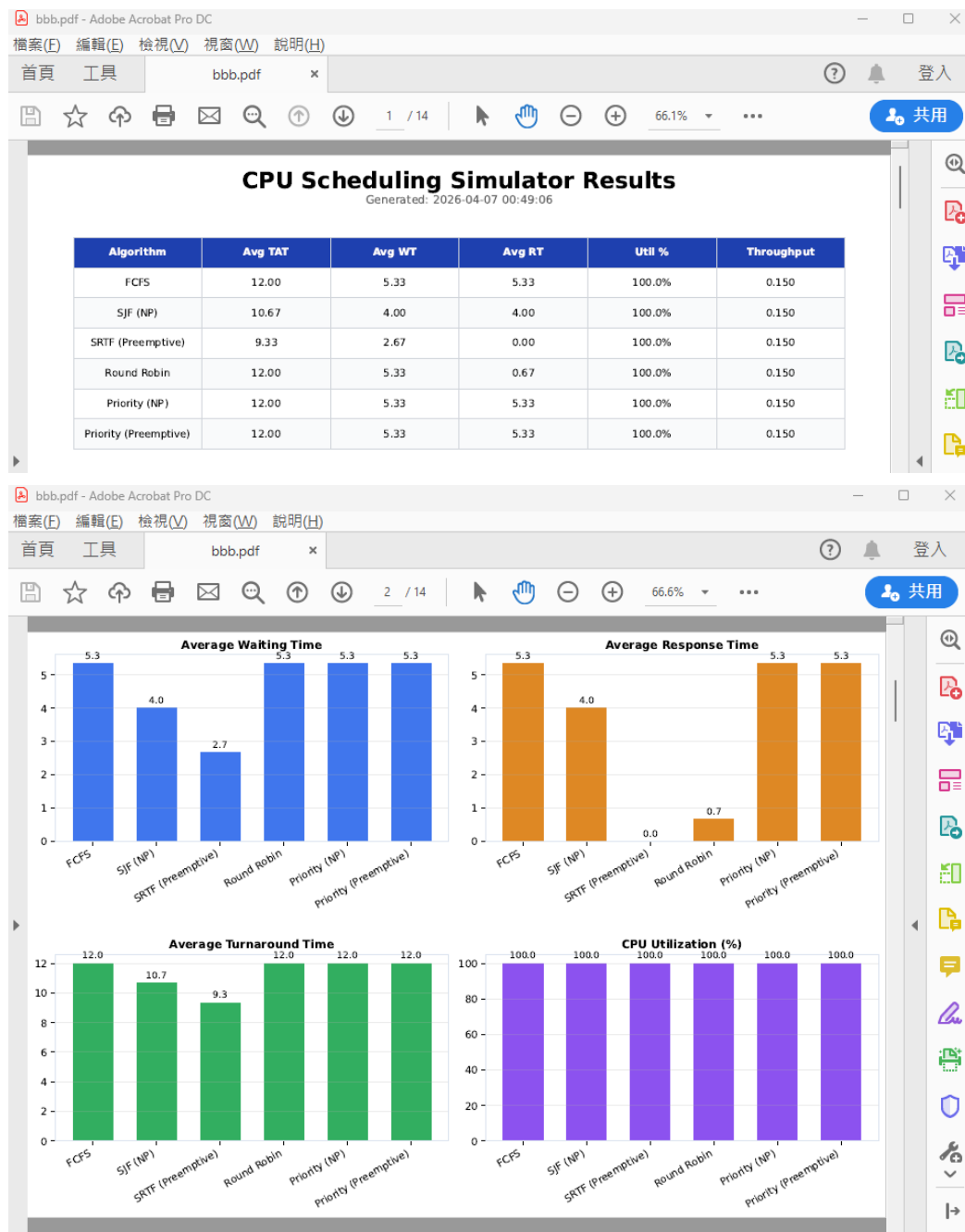


- (iii) Once saved, a pop-up message with the indication of file location will be shown.



- (iv) The PDF contains:
- (a) One Summary sheet (comparison table);
 - (b) One Comparison chart page

(c) Six * 2-page algorithm sheets (process results table + Gantt chart)



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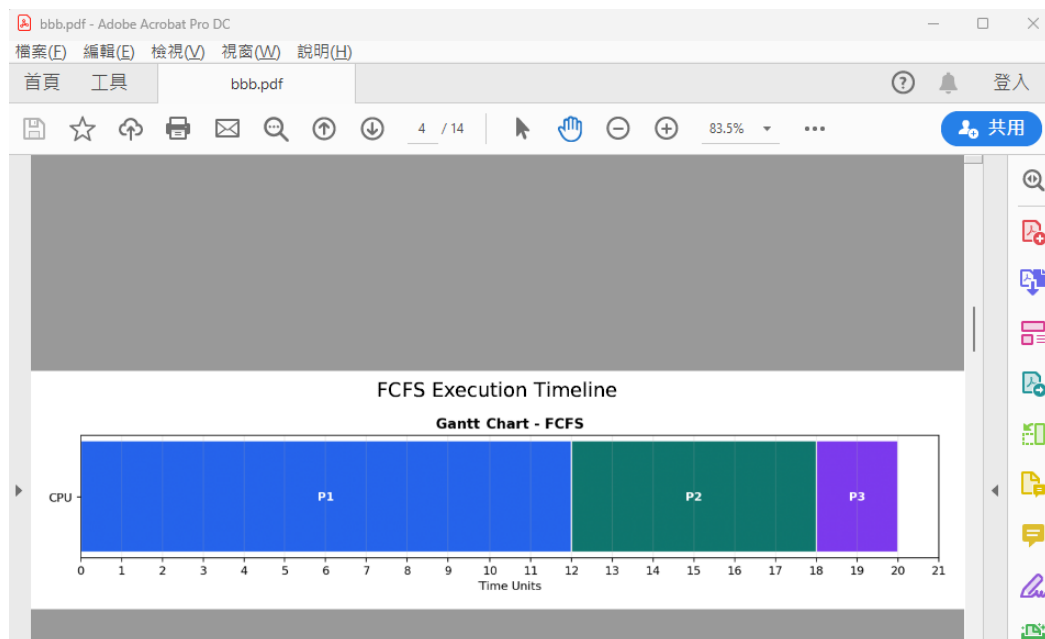
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共用

Detailed Analysis: FCFS

PID	Arrival	Burst	Prio	Start	Comp	TAT	WT	RT
1	0.0	12	0	0.0	12.0	12.0	0.0	0.0
2	4.0	6	0	12.0	18.0	14.0	8.0	8.0
3	10.0	2	0	18.0	20.0	10.0	8.0	8.0
AVG						12.00	5.33	5.33



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共用

Detailed Analysis: Round Robin

PID	Arrival	Burst	Prio	Start	Comp	TAT	WT	RT
1	0.0	12	0	0.0	20.0	20.0	8.0	0.0
2	4.0	6	0	4.0	16.0	12.0	6.0	0.0
3	10.0	2	0	12.0	14.0	4.0	2.0	2.0
AVG						12.00	5.33	0.67